

IMEN266 Operations Research II

Concept & Proof Companion — Chapter 5 (Exponential distribution, Poisson process, CTMCs)

[FULL version — all blanks filled]

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How to use this document

This companion *replaces your class notes*: every definition and theorem from the Ch. 5 slides (and the derivations developed in class) is written out here in full, so you can watch and think in class instead of transcribing. It does **not** replace *doing*: for each core proof, (1) work through the GAP-FILL version with pen and paper, (2) check against the FULL version, (3) answer the Checkpoints, and (4) if stuck — or done — use an AI study card. Exams are closed-book; these eight proofs and their checkpoints are exactly the kind of reasoning the exams reward. Items marked **(midterm-style)** or **(final-style)** mirror questions from past exams: proving memorylessness (Proof 1), modeling a queue with reneging and balking as a CTMC (Proof 4), the limiting distribution of a birth & death system such as $M/M/2/K$ (Proof 7), and a long-run cost (Proof 8).

1 Concept digest

1.1 The exponential distribution (slides pp. 1–3/63)

$X \sim \exp(\lambda)$, $\lambda > 0$, has density and cdf

$$f(x) = \lambda e^{-\lambda x} \ (x \geq 0), \quad F(x) = P(X \leq x) = 1 - e^{-\lambda x}, \quad P(X > x) = e^{-\lambda x},$$

$$E[X] = \frac{1}{\lambda}, \quad \text{Var}[X] = \frac{1}{\lambda^2}.$$

λ is a **rate** (events per unit time); always convert times to the unit of λ before computing. Three facts drive the whole chapter:

1. **Memoryless:** $P(X > t + s \mid X > s) = P(X > t)$ — a used component is as good as a new one; the exponential is the *only* continuous distribution with this property (Proof 1).
2. **Minimum:** if $X_i \sim \exp(\lambda_i)$ are independent, then $\min_i X_i \sim \exp(\lambda_1 + \cdots + \lambda_n)$ — the *rates* add, not the means (Proof 1).
3. **Which clock rings first:** $P(X_i = \min_j X_j) = \lambda_i / \sum_j \lambda_j$, independently of *when* the minimum occurs (Proof 1).

The sum of n iid $\exp(\lambda)$ variables is **Erlang**(n, λ) (a Gamma with integer shape), mean n/λ , variance n/λ^2 — more regular than a single exponential with the same mean. (*verified in notebook: CP6 warm-up, #1, #2*)

1.2 The Poisson process (slides pp. 9–12/63)

Let $X_1, X_2, \dots$ be iid $\exp(\lambda)$, $S_n = X_1 + \dots + X_n$ the time of the n -th event, and $N(t) = \max\{n : S_n \leq t\}$ the number of events in $(0, t]$. Then $\{N(t), t \geq 0\}$ is a **Poisson process with rate λ** , $\text{PP}(\lambda)$:

$$P(N(t) = n) = e^{-\lambda t} \frac{(\lambda t)^n}{n!}, \quad E[N(t)] = \text{Var}[N(t)] = \lambda t \quad (\text{Proof 2}),$$

with **independent increments** (counts over disjoint intervals are independent) and **stationary increments** (the count over an interval depends only on its length) — both consequences of the memoryless gaps. Key identity: $\{N(t) \geq n\} = \{S_n \leq t\}$.

Splitting (thinning) and merging (superposition) (Proof 3). If each event of a $\text{PP}(\lambda)$ is independently sent to branch i with probability p_i , branch i is a $\text{PP}(\lambda p_i)$ and the branches are independent. If independent $\text{PP}(\lambda_1), \dots, \text{PP}(\lambda_n)$ are merged, the result is $\text{PP}(\lambda_1 + \dots + \lambda_n)$. **Round-robin** routing (every k -th event to a branch) is *not* splitting: the branch gaps are $\text{Erlang}(k, \lambda)$, not exponential, and the branches are not independent. (*verified in notebook: CP6 #3 and its thinning-vs-round-robin experiment*)

1.3 Continuous-time Markov chains (slides pp. 16–18/63)

$\{X(t), t \geq 0\}$ with countable state space $\mathcal{S}$ is a **CTMC** if for all $s, t \geq 0$ and states,

$$P(X(t+s) = j \mid X(s) = i, X(u) = x(u) \text{ for } 0 \leq u < s) \stackrel{(1)}{=} P(X(t+s) = j \mid X(s) = i) \\ \stackrel{(2)}{=} P(X(t) = j \mid X(0) = i) =: p_{ij}(t),$$

(1) the **Markov property** (the history before s is irrelevant once $X(s)$ is known) and (2) **time homogeneity** (only the elapsed time matters). The Markov property forces the time spent in a state to be exponential (Proof 4): in state i , one independent $\exp(q_{ij})$ clock runs for each possible transition $i \rightarrow j$; the sojourn time is $\exp(\lambda_i)$ with $\lambda_i = \sum_{j \neq i} q_{ij}$, and the chain jumps to j with probability q_{ij}/λ_i . The **generator** (rate matrix) is

$$Q = [q_{ij}], \quad q_{ij} = \text{rate of the clock } i \rightarrow j \ (j \neq i), \quad q_{ii} = -\sum_{j \neq i} q_{ij} = -\lambda_i,$$

so every row of Q sums to 0. The **rate diagram** draws states as nodes and $q_{ij} > 0$ as arrows labelled by rates — *no self-loops*: $p_{ii}(t)$ includes leaving and returning.

Modeling recipe (slide p. 16/63; every CP7 problem): (1) define $X(t)$; (2) write $\mathcal{S}$; (3) check the Markov property — every duration is exponential and independent of the past; (4) derive Q by counting the clocks running in each state (k machines up $\Rightarrow$ failure rate $k\mu$), and check the row sums; (5) draw the diagram. If a natural state hides the future (who is being repaired under FCFS), *enlarge the state* until it is Markov (CP7 #4). A **birth & death** process is a CTMC on $\{0, 1, 2, \dots\}$ with only ± 1 jumps, rates λ_k up and μ_k down. (*verified in notebook: CP7 #1–#5, CTMC(Q , states) row-sum check*)

1.4 Transient analysis (slides pp. 29–32/63)

$P(t) = [p_{ij}(t)]$ is stochastic for each t , $P(0) = I$, and satisfies the **semigroup (Chapman–Kolmogorov) property** $P(t+s) = P(t)P(s)$ (Proof 5). Differentiating gives the **Kolmogorov forward equations**

$$\frac{d}{dt}P(t) = P(t)Q, \quad Q = \lim_{h \downarrow 0} \frac{P(h) - I}{h}, \quad \text{solved by} \quad P(t) = e^{Qt} = \sum_{n \geq 0} \frac{(Qt)^n}{n!}$$

(the backward equations $\frac{d}{dt}P(t) = QP(t)$ hold too, since $P(t)$ commutes with Q). For a row vector $p(t) = [P(X(t) = j)]_j$: $p(t) = p(0)e^{Qt}$ and $\frac{d}{dt}p(t) = p(t)Q$. In practice e^{Qt} is computed by `scipy.linalg.expm`; the truncated series and $(I + Qt/N)^N$ converge too (CP8 #8 shows how slowly). Two-state closed form: Proof 5. (*verified in notebook: CP8 #1, #8*)

1.5 Embedded chain, ergodicity, limiting distribution (slides pp. 33–39, 46/63)

Let S_n be the time of the n -th jump and $X_n = X(S_n)$. The **embedded Markov chain** $\{X_n\}$ is a DTMC with $P_{ij} = q_{ij}/\lambda_i$ ($j \neq i$), $P_{ii} = 0$ (Proof 4). Irreducibility ($p_{ij}(t) > 0$ for all i, j and some $t > 0$) is the same for the CTMC and its embedded chain (same arrows); positive recurrence is *not* equivalent in general: with π the stationary distribution of the embedded chain, the CTMC is positive recurrent iff $\sum_j \pi_j/\lambda_j < \infty$ (finite $\mathcal{S}$: automatic). There is **no aperiodicity condition** in continuous time — sojourn times are continuous, so nothing can cycle.

Theorem. If the CTMC is irreducible and positive recurrent (**ergodic**), then the limit $p_j = \lim_{t \rightarrow \infty} P(X(t) = j)$ exists, does not depend on the initial state, equals the long-run fraction of time spent in j , and $\mathbf{p} = (p_j)$ is the **unique** solution of

$$\mathbf{p}Q = 0, \quad \sum_{j \in \mathcal{S}} p_j = 1 \quad (\text{Proof 6}).$$

Rearranged, $\mathbf{p}Q = 0$ says for each state j : (rate out of j) $\times p_j = \sum_{i \neq j} p_i q_{ij}$ — the **node balance equations**. For birth & death processes they reduce to the **arc balance equations** $\lambda_k p_k = \mu_{k+1} p_{k+1}$ with the product-form solution (Proof 7). Three ways to solve (slide p. 46/63): expand and solve; node balance; arc balance (birth & death, or any time-reversible chain).

p versus π . π_j is the long-run fraction of *jumps* into j , p_j the fraction of *time* in j ; they differ unless all λ_j are equal, and $p_j \propto \pi_j/\lambda_j$ (Proof 6). Example: machine up/down has $\pi = (\frac{1}{2}, \frac{1}{2})$ but $\mathbf{p} = (\frac{\mu}{\lambda+\mu}, \frac{\lambda}{\lambda+\mu})$. (*verified in notebook: CP8 warm-up, #1–#4*)

1.6 Cost and reward (slide p. 59/63)

If cost accrues at rate $c(j)$ per unit time while the chain is in j , the **long-run average cost per unit time** is

$$C = \lim_{T \rightarrow \infty} \frac{1}{T} \int_0^T c(X(t)) dt = \sum_{j \in \mathcal{S}} c(j) p_j \quad (\text{Proof 8}).$$

Time fractions, not jump fractions: use $\mathbf{p}$, never π . (*verified in notebook: CP8 #8, #9*)

1.7 Chain zoo (the class-problem models)

model (CP7/CP8)	state $X(t)$	$ \mathcal{S} $	type	limiting distribution
machine up/down (#1)	0/1	2	B&D	$\mathbf{p} = (\mu, \lambda)/(\lambda + \mu)$; $P(t)$ in closed form (Proof 5)
two photocopy machines (#2)	# up	3	B&D	$(1, 10, 25)/36 = \text{binomial}(2, \frac{5}{6})$
switch with K channels (#3)	# calls	$K+1$	B&D	$p_k \propto (\lambda/\mu)^k/k!$; $p_K = \text{Erlang loss}$ (Proof 7)
amoebae (#5)	# alive	∞	B&D	positive recurrent iff births are slow enough; truncate in code
two machines, FCFS (#4)	who is up / in repair	5	general	node balance; closed form with $D = \lambda_1\mu_1 + \lambda_1\lambda_2 + \lambda_2\mu_2$
light bulbs (CP8 #5)	# alive	$B+1$	pure death	absorbed at 0; gaps $\exp((B-k)/\theta)$
barbershop (CP8 #6)	(x_1, x_2, x_3)	8	general	numeric; $p_{111} = \text{fraction of lost customers}$
blanks (CP8 #8)	0/1/2	3	general	$(2, 4, 3)/9$; $P(1)_{02} \approx 0.371$
repair persons (CP8 #9)	# up	3	B&D	two persons iff $r/\mu > c/\lambda$ (Proof 8)

2 Core proofs

Notation: $\mathcal{S}$ finite unless stated; $\lambda_i = -q_{ii}$; $\mathbf{p}$ is the limiting (time-fraction) distribution and π the stationary distribution of the embedded chain. Blanks are the steps *you* must supply in the gap-fill version.

Proof 1 — Memorylessness, the minimum, and which clock rings first (midterm-style)

Theorem. Let $X \sim \exp(\lambda)$. (a) $P(X > t + s \mid X > s) = P(X > t)$ for all $s, t \geq 0$. (b) Conversely, if X is a continuous random variable on $(0, \infty)$ with this property, then X is exponential. (c) If $X_1, \dots, X_n$ are independent, $X_i \sim \exp(\lambda_i)$, then $Z = \min_i X_i \sim \exp(\lambda_1 + \dots + \lambda_n)$ and $P(X_i = Z) = \lambda_i / \sum_j \lambda_j$.

Proof. (a) Since $\{X > t + s\} \subset \{X > s\}$,

$$P(X > t + s \mid X > s) = \frac{P(X > t + s, X > s)}{P(X > s)} = \frac{P(X > t + s)}{P(X > s)} = \frac{e^{-\lambda(t+s)}}{e^{-\lambda s}} = e^{-\lambda t} = P(X > t).$$

(b) Write $g(t) = P(X > t)$. Memorylessness says $g(t + s) = g(t)g(s)$ for all $s, t \geq 0$. Hence $g(2/n) = g(1/n)^2$, and by induction $g(m/n) = g(1/n)^m$ and $g(1) = g(1/n)^n$, so $g(m/n) = g(1)^{m/n}$ for all rationals. g is monotone, so $g(t) = g(1)^t = e^{-\lambda t}$ with $\lambda = -\ln g(1) \in (0, \infty)$ (since $0 < g(1) < 1$ for a continuous X on $(0, \infty)$).

(c) $P(Z > z) = P(\text{all } X_i > z) = \prod_i P(X_i > z)$ by [independence](#), which is $\prod_i e^{-\lambda_i z} = e^{-(\sum_i \lambda_i)z}$. For the second claim take $n = 2$ and condition on X_2 :

$$P(X_1 < X_2) = \int_0^\infty P(X_1 < y) f_{X_2}(y) dy = \int_0^\infty (1 - e^{-\lambda_1 y}) \lambda_2 e^{-\lambda_2 y} dy = 1 - \frac{\lambda_2}{\lambda_1 + \lambda_2} = \frac{\lambda_1}{\lambda_1 + \lambda_2}.$$

For general n apply this with X_2 replaced by $\min_{j \neq i} X_j \sim \exp(\sum_{j \neq i} \lambda_j)$. Moreover the event $\{X_i = Z\}$ is independent of Z : $P(X_i = Z, Z > z) = P(X_i > z, X_j > X_i \forall j \neq i)$, which by the

same computation from z on (memorylessness of every clock) equals $e^{-(\sum_j \lambda_j)z} \cdot \lambda_i / \sum_j \lambda_j = P(Z > z)P(X_i = Z)$. $\square$

(verified in notebook: CP6 warm-up (residual life histograms), CP7 warm-up (winner frequencies))

Checkpoint — answer before moving on.

- **(midterm-style)** Prove (a) from scratch in three lines, stating where you use $\{X > t + s\} \subset \{X > s\}$.
- Four welders at rate 3/h; none has finished by 10:20 (CP6 #2(d)). Why is the standard deviation of the remaining time exactly 1/12 h, whatever the elapsed times were?
- In (c), which step fails if the clocks are *dependent*? Give a two-clock example where $\min(X_1, X_2)$ is not exponential.
- A discrete analogue: which distribution on $\{1, 2, \dots\}$ is memoryless ($P(X > m + n \mid X > n) = P(X > m)$)? Prove it along the lines of (b).

AI study card — paste one of these into your AI assistant:

- Quiz me one step at a time on the proof that exponential is the only memoryless continuous distribution. Make me derive $g(m/n) = g(1)^{(m/n)}$ and say where monotonicity is used.
- Give me three independent exponential clocks with rates you choose; ask me for the distribution of the first ring and the probability each clock wins. Then ask whether 'who wins' and 'when' are independent, and check my reasoning.

Proof 2 — Poisson counts from exponential gaps

Theorem. If the gaps $X_1, X_2, \dots$ are iid $\exp(\lambda)$, then $N(t) \sim \text{Poisson}(\lambda t)$ for every $t > 0$.

Proof. *Step 1:* $S_n = X_1 + \dots + X_n$ is Erlang. By induction, with the convolution $f_{S_n}(s) = \int_0^s f_{S_{n-1}}(u) \lambda e^{-\lambda(s-u)} du$, one gets

$$f_{S_n}(s) = \lambda e^{-\lambda s} \frac{(\lambda s)^{n-1}}{(n-1)!} \quad (s \geq 0) : \quad n \rightarrow n+1 : \quad \int_0^s \lambda e^{-\lambda u} \frac{(\lambda u)^{n-1}}{(n-1)!} \lambda e^{-\lambda(s-u)} du = \lambda e^{-\lambda s} \frac{(\lambda s)^n}{n!}.$$

Step 2: counts from arrival times. Since $\{N(t) \geq n\} = \{S_n \leq t\}$,

$$P(N(t) = n) = P(S_n \leq t) - P(S_{n+1} \leq t) = \int_0^t f_{S_n}(s) ds - \int_0^t f_{S_{n+1}}(s) ds.$$

Integrating $f_{S_{n+1}}$ by parts, $\int_0^t \lambda e^{-\lambda s} \frac{(\lambda s)^n}{n!} ds = \left[-e^{-\lambda s} \frac{(\lambda s)^n}{n!} \right]_0^t + \int_0^t e^{-\lambda s} \frac{\lambda(\lambda s)^{n-1}}{(n-1)!} ds = -e^{-\lambda t} \frac{(\lambda t)^n}{n!} + P(S_n \leq t)$, so the difference is $P(N(t) = n) = e^{-\lambda t} \frac{(\lambda t)^n}{n!}$. $\square$

Remark. Independent and stationary increments follow from the memoryless gaps: at any time t the residual gap is again $\exp(\lambda)$, independent of the past, so the process restarts. Conversely (Ross, Thm. 5.1) a counting process with stationary independent increments and $P(N(h) = 1) = \lambda h + o(h)$, $P(N(h) \geq 2) = o(h)$ is a PP(λ).

(verified in notebook: CP6 #3(a)–(e): window counts of a simulated PP(10))

Checkpoint — answer before moving on.

- Where exactly is $\{N(t) \geq n\} = \{S_n \leq t\}$ used, and why is it “ $\geq$ ” and not “ $=$ ”?

- Compute $E[N(t)]$ and $\text{Var}[N(t)]$ from the pmf. Why does $\text{var} = \text{mean}$ fail for the light-bulb process of CP8 #5?
- The 4-lane traffic at East Gate (CP6 #3): the count on the left lane over 15 s is Poisson with which parameter? Two different justifications.

AI study card — paste one of these into your AI assistant:

- Walk me through Step 2 of the Erlang-to-Poisson proof by asking questions: why the difference of two cdfs, and what integration by parts produces. Do not show me the result until I have tried.
- Give me a counting process whose gaps are iid but uniform on $[0, 2]$. Ask me why it is not Poisson and to compute $P(N(1)=0)$. Check my answer.

Proof 3 — Thinning and superposition

Theorem. (a) If each event of a $\text{PP}(\lambda)$ is independently of type 1 with probability p and type 2 otherwise, the type-1 and type-2 processes are independent Poisson processes with rates λp and $\lambda(1-p)$. (b) If $N_1 \sim \text{PP}(\lambda_1)$ and $N_2 \sim \text{PP}(\lambda_2)$ are independent, then $N_1 + N_2 \sim \text{PP}(\lambda_1 + \lambda_2)$.

Proof. (a) Fix t and condition on the total count:

$$\begin{aligned} P(N_1(t) = n, N_2(t) = m) &= P(N(t) = n + m) \binom{n+m}{n} p^n (1-p)^m \\ &= e^{-\lambda t} \frac{(\lambda t)^{n+m}}{(n+m)!} \cdot \frac{(n+m)!}{n! m!} p^n (1-p)^m \\ &= e^{-\lambda p t} \frac{(\lambda p t)^n}{n!} \cdot e^{-\lambda(1-p)t} \frac{(\lambda(1-p)t)^m}{m!}, \end{aligned}$$

using $e^{-\lambda t} = e^{-\lambda p t} e^{-\lambda(1-p)t}$. The joint pmf factorises into $\text{Poisson}(\lambda p t) \times \text{Poisson}(\lambda(1-p)t)$, so $N_1(t)$ and $N_2(t)$ are independent Poisson variables; the same computation over disjoint intervals (independent increments of N , independent type choices) gives independent stationary increments for each thinned process, so both are Poisson processes and they are independent of each other.

(b) The gap until the next event of the merged process is the minimum of two independent residual gaps, $\exp(\lambda_1)$ and $\exp(\lambda_2)$ (memoryless), hence $\exp(\lambda_1 + \lambda_2)$; by memorylessness the merged process restarts at each event with the same structure, so its gaps are iid $\exp(\lambda_1 + \lambda_2)$. Alternatively, $N_1(t) + N_2(t)$ is a sum of independent Poissons, which is $\text{Poisson}((\lambda_1 + \lambda_2)t)$, with independent stationary increments. $\square$

(verified in notebook: CP6 #3(f),(g); the thinning-vs-round-robin experiment)

Checkpoint — answer before moving on.

- In (a), which two independence assumptions are used, and which one round-robin routing violates?
- Trucks $\text{PP}(8)$ and buses $\text{PP}(2)$: given that a vehicle just passed, what is the probability it was a bus? Which part of the proof answers this?
- Every second car goes left. Compute the mean and variance of the left-lane gap and compare with $\text{PP}(\lambda/2)$. Which is more regular?

AI study card — paste one of these into your AI assistant:

- Ask me to prove that thinning a Poisson process with probability p gives a Poisson process of rate λp , by conditioning on $N(t)$. Stop me if I forget the binomial factor.
- Explain to me, then quiz me, why merging two independent Poisson processes gives a Poisson process, using the minimum of exponentials. Then ask what changes if the two processes are NOT independent.

Proof 4 — Exponential sojourns, the embedded chain, and the jump probabilities (final-style)

Theorem. Let $\{X(t)\}$ be a CTMC and T_i the time it stays in state i after entering it. (a) $T_i \sim \exp(\lambda_i)$ for some $\lambda_i \in [0, \infty)$. (b) Conversely, a process that stays in i for an $\exp(\lambda_i)$ time and then jumps to j with probability P_{ij} (independently of the past and of the sojourn) is a CTMC with $q_{ij} = \lambda_i P_{ij}$. (c) The jump sequence $X_n = X(S_n)$ is a DTMC with $P_{ij} = q_{ij}/\lambda_i$, $P_{ii} = 0$.

Proof. (a) For $s, t \geq 0$, the event $\{T_i > s\}$ is “ $X(u) = i$ for $0 \leq u \leq s$ ”. By the Markov property at time s and time homogeneity,

$$\begin{aligned} P(T_i > t + s \mid T_i > s) &= P(X(u) = i \ \forall u \leq t + s \mid X(u) = i \ \forall u \leq s) \\ &= P(X(u) = i \ \forall u \leq t \mid X(0) = i) = P(T_i > t), \end{aligned}$$

so T_i is memoryless; being continuous (or identically ∞), it is exponential by Proof 1(b), with $\lambda_i = 0$ for an absorbing state.

(b) Given $X(s) = i$, the future is determined by the residual sojourn $\exp(\lambda_i)$ (memoryless, independent of the past) and by the subsequent jumps and sojourns, none of which look at the history: this is the Markov property and, since nothing depends on s , time homogeneity. For the rates: in a short interval h , $P(X(h) = j \mid X(0) = i) = P(\text{jump before } h) P_{ij} + o(h) = (1 - e^{-\lambda_i h}) P_{ij} + o(h) = \lambda_i P_{ij} h + o(h)$, so $q_{ij} = \lambda_i P_{ij}$ and $q_{ii} = -\sum_{j \neq i} q_{ij} = -\lambda_i$.

(c) Conversely, from the clock description: in state i the independent clocks $\exp(q_{ij})$, $j \neq i$, compete; the first to ring is $\exp(\sum_{j \neq i} q_{ij}) = \exp(\lambda_i)$ and clock j wins with probability q_{ij}/λ_i , independently of when (Proof 1(c)). Given the jump chain and the sojourns are independent, the sequence of states visited is a DTMC with these probabilities. $\square$

(verified in notebook: CP7 warm-up; CTMC.embedded.chain(); CP8 #1 ($\pi \neq p$))

Checkpoint — answer before moving on.

- **(final-style)** A single-server queue: Poisson arrivals at rate λ , $\exp(\mu)$ service, a waiting customer abandons after an $\exp(\theta)$ patience time (*reneging*), and an arrival that finds i customers joins only with probability $1/(i+1)$ (*balking*). Define the state, give $q_{i,i+1}$ and $q_{i,i-1}$, and draw the rate diagram. Which clocks run in state i ?
- Deterministic service times of 20 minutes: what is $P(T_i > t + s \mid T_i > s)$, and which line of (a) fails?
- The one-machine chain has embedded matrix $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ for every λ, μ . What information about the CTMC is lost in the embedded chain, and how is it recovered in Proof 6?

AI study card — paste one of these into your AI assistant:

- Quiz me on why the time spent in a state of a CTMC must be exponential. Make me write the event $\{T_i > s\}$ in terms of the path and apply the Markov property.
- Give me a queue with reneging and balking (rates you choose) and ask me for the generator on states 0..4. Then check each entry and ask me which exponential clock produces it.

Proof 5 — Chapman–Kolmogorov, the forward equations, and $P(t) = e^{Qt}$

Theorem. (a) $P(t+s) = P(t)P(s)$. (b) $\frac{d}{dt}P(t) = P(t)Q$ with $Q = \lim_{h \downarrow 0}(P(h) - I)/h$; hence $P(t) = e^{Qt}$. (c) For the two-state chain $Q = \begin{pmatrix} -\lambda & \lambda \\ \mu & -\mu \end{pmatrix}$,

$$P(t) = \frac{1}{\lambda + \mu} \begin{pmatrix} \mu & \lambda \\ \mu & \lambda \end{pmatrix} + \frac{e^{-(\lambda+\mu)t}}{\lambda + \mu} \begin{pmatrix} \lambda & -\lambda \\ -\mu & \mu \end{pmatrix}.$$

Proof. (a) Condition on the state at time t :

$$p_{ij}(t+s) = \sum_{k \in S} P(X(t+s) = j \mid X(t) = k, X(0) = i) P(X(t) = k \mid X(0) = i) = \sum_k p_{kj}(s) p_{ik}(t),$$

by the Markov property (drop $X(0) = i$) and time homogeneity (shift $(t, t+s]$ to $(0, s]$). (b) $\frac{P(t+h)-P(t)}{h} = \frac{P(t)P(h)-P(t)}{h} = P(t)\frac{P(h)-I}{h} \rightarrow P(t)Q$. The entries of Q : for $j \neq i$, $p_{ij}(h) = q_{ij}h + o(h)$ (one jump $i \rightarrow j$ in $(0, h]$, Proof 4(b)), so $q_{ij} \geq 0$ is the rate $i \rightarrow j$; and $p_{ii}(h) - 1 = -\sum_{j \neq i} p_{ij}(h)$ gives $q_{ii} = -\sum_{j \neq i} q_{ij}$. The matrix ODE $P' = PQ$, $P(0) = I$, has the unique solution $P(t) = e^{Qt} := \sum_n (Qt)^n/n!$: differentiate the series term by term, $\frac{d}{dt} \sum_n \frac{Q^n t^n}{n!} = \sum_{n \geq 1} \frac{Q^n t^{n-1}}{(n-1)!} = \left(\sum_m \frac{Q^m t^m}{m!} \right) Q$.

(c) For $P(t) = A + e^{-(\lambda+\mu)t}B$ with the two matrices displayed: $P(0) = A + B = I$, and $P'(t) = -(\lambda + \mu)e^{-(\lambda+\mu)t}B$ while $P(t)Q = AQ + e^{-(\lambda+\mu)t}BQ$. Now $AQ = 0$ (each row of A is $(\mu, \lambda)/(\lambda + \mu)$ and $(\mu \ \lambda)Q = 0$) and $BQ = \frac{1}{\lambda+\mu} \begin{pmatrix} \lambda & -\lambda \\ -\mu & \mu \end{pmatrix} \begin{pmatrix} -\lambda & \lambda \\ \mu & -\mu \end{pmatrix} = -(\lambda + \mu)B$, so $P' = PQ$ and the formula solves the forward equations; by uniqueness it is e^{Qt} . $\square$

Remark. $P(t) \rightarrow A$ as $t \rightarrow \infty$ at rate $\lambda + \mu$ (the nonzero eigenvalue of Q is $-(\lambda + \mu)$): every row of $P(\infty)$ is $\mathbf{p} = (\mu, \lambda)/(\lambda + \mu)$ — the start is forgotten. This is the continuous-time twin of Ch. 4, Proof 4.

(verified in notebook: CP8 #1 (both rows of $P(t)$ converge), CP8 #8 (expm, series, $(I+Q/N)^N$, solve_ivp agree))

Checkpoint — answer before moving on.

- Write the *backward* equations by conditioning on the first jump instead of the last instant. Why do forward and backward equations have the same solution?
- For CP8 #8 the truncated series needs 34 terms at $t = 1$ although $\|Q\| = 5$. Explain with $5^n/n!$, and predict the number of terms for $t = 3$.
- From (c), find $P(X(t) = \text{up} \mid X(0) = \text{down})$ and the time at which it reaches 95% of its limit. Which quantity of the chain sets that time scale?

AI study card — paste one of these into your AI assistant:

- Ask me to derive the Kolmogorov forward equation from the semigroup property, one limit at a time, and to identify each entry of Q as a rate. Do not let me skip why q_{ii} is negative.
- Give me a 2-state generator with numbers and make me verify the closed form of $P(t)$ by checking $P(0)=I$ and $P'=PQ$. Then ask what $P(t)$ tends to and how fast.

Proof 6 — The limiting distribution: $pQ = 0$, time fractions, and $p \neq \pi$

Theorem. Let the CTMC be irreducible with finite $\mathcal{S}$. Then (a) the system $pQ = 0$, $\sum_j p_j = 1$ has a unique solution with all $p_j > 0$; (b) $P(t) \rightarrow \mathbf{1}p$ as $t \rightarrow \infty$ (every row tends to p), so $p_j = \lim_t P(X(t) = j)$ for any initial distribution; (c) p_j is the long-run fraction of time in j ; (d) $p_j \propto \pi_j/\lambda_j$, where π is the stationary distribution of the embedded chain.

Proof. (a) The columns of Q sum to the zero vector row-wise: $Q\mathbf{1} = 0$, so $\text{rank } Q \leq |\mathcal{S}| - 1$; for an irreducible generator the rank is exactly $|\mathcal{S}| - 1$ (uniformization, below, reduces this to the DTMC statement of Ch. 4), so the solution space of $pQ = 0$ is one-dimensional and the normalisation picks one vector.

(b) *Uniformization.* Let $\Lambda \geq \max_i \lambda_i$ and $\tilde{P} = I + Q/\Lambda$: a stochastic matrix (entries ≥ 0 , rows sum to 1) with $\tilde{P}_{ii} \geq 1 - \lambda_i/\Lambda \geq 0$, so the DTMC with matrix $\tilde{P}$ is irreducible and, if $\Lambda > \max_i \lambda_i$, aperiodic (self-loops). Then $Q = \Lambda(\tilde{P} - I)$ and

$$P(t) = e^{Qt} = e^{-\Lambda t} e^{\Lambda t \tilde{P}} = \sum_{n \geq 0} e^{-\Lambda t} \frac{(\Lambda t)^n}{n!} \tilde{P}^n :$$

the CTMC is the DTMC $\tilde{P}$ observed at the events of a PP(Λ) (fictitious self-transitions). By Ch. 4, $\tilde{P}^n \rightarrow \mathbf{1}\tilde{\pi}$ with $\tilde{\pi}\tilde{P} = \tilde{\pi}$, and $\tilde{\pi}\tilde{P} = \tilde{\pi} \iff \tilde{\pi}Q = 0$, so $\tilde{\pi} = p$; since the Poisson weights put all their mass on large n as $t \rightarrow \infty$, $P(t) \rightarrow \mathbf{1}p$. Directly from the forward equations: if $P(t)$ converges, then $P'(t) \rightarrow 0$, so $P(\infty)Q = 0$ and each row of $P(\infty)$ solves $pQ = 0$.

(c) Let $V_j(T) = \int_0^T \mathbf{1}\{X(t) = j\} dt$. Taking expectations, $E[V_j(T)]/T = \frac{1}{T} \int_0^T P(X(t) = j) dt \rightarrow p_j$ (Cesàro mean of a convergent function). That $V_j(T)/T$ itself converges (almost surely) is the renewal-reward theorem of Ch. 7: successive visits to j form cycles of mean length m_j , each contributing an $\exp(\lambda_j)$ sojourn, so $V_j(T)/T \rightarrow \frac{1/\lambda_j}{m_j}$, and consistency with (b) gives $p_j = 1/(\lambda_j m_j)$.

(d) Let $P_{ij} = q_{ij}/\lambda_i$ ($j \neq i$), $P_{ii} = 0$, and $\pi = \pi P$. Put $p_j = c \pi_j/\lambda_j$. Then for each j ,

$$(pQ)_j = \sum_{i \neq j} p_i q_{ij} + p_j q_{jj} = c \left(\sum_{i \neq j} \frac{\pi_i}{\lambda_i} \lambda_i P_{ij} - \frac{\pi_j}{\lambda_j} \lambda_j \right) = c \left(\sum_{i \neq j} \pi_i P_{ij} - \pi_j \right) = 0,$$

since $\sum_{i \neq j} \pi_i P_{ij} = (\pi P)_j = \pi_j$. By uniqueness in (a), p is this vector normalised: $p_j = \frac{\pi_j/\lambda_j}{\sum_k \pi_k/\lambda_k}$.

□

Reading $pQ = 0$. Component j says $\sum_{i \neq j} p_i q_{ij} = \lambda_j p_j$: in the long run, the rate of transitions *into* j equals the rate *out of* j — node balance. On a countable $\mathcal{S}$ the theorem needs positive recurrence, i.e. $\sum_j \pi_j/\lambda_j < \infty$, which is exactly the normalisability in (d).

(verified in notebook: CP8 warm-up (three ways), #1 (time vs jump fractions), #2, #4)

Checkpoint — answer before moving on.

- CP8 #1: $\pi = (\frac{1}{2}, \frac{1}{2})$ but $p = (\mu, \lambda)/(\lambda + \mu)$. Verify (d) by hand. Which state is over-represented in time, and why?

- Explain in one sentence why no aperiodicity assumption appears, using the uniformized chain $\tilde{P}$ with $\Lambda > \max_i \lambda_i$.
- **(final-style)** A 3-state CTMC has Q with rows $(-3, 1, 2)$, $(2, -2, 0)$, $(1, 1, -2)$. Write the three node balance equations, drop the redundant one, and solve for $\mathbf{p}$. Then compute π of the embedded chain and check (d).

AI study card — paste one of these into your AI assistant:

- Quiz me on why the limiting distribution of a CTMC solves $\mathbf{p}Q = 0$ and NOT $\pi_i = \pi_i$ of the embedded chain. Give me a 2-state example with very different sojourn rates and make me compute both.
- Explain uniformization to me with a 3-state example, then ask me why the uniformized DTMC is aperiodic and what that implies about periodicity for CTMCs.

Proof 7 — Birth & death processes: arc balance, product form, Erlang loss (final-style)

Theorem. For an irreducible birth & death process on $\{0, 1, \dots, K\}$ with birth rates λ_k ($k < K$) and death rates μ_k ($k \geq 1$):

$$\lambda_k p_k = \mu_{k+1} p_{k+1} \quad (0 \leq k < K), \quad p_k = p_0 \prod_{i=1}^k \frac{\lambda_{i-1}}{\mu_i}, \quad p_0 = \left(\sum_{k=0}^K \prod_{i=1}^k \frac{\lambda_{i-1}}{\mu_i} \right)^{-1}.$$

For the K -channel switch ($\lambda_k = \lambda$, $\mu_k = k\mu$), with $a = \lambda/\mu$: $p_k = \frac{a^k/k!}{\sum_{i=0}^K a^i/i!}$, and p_K is the **Erlang loss formula** $B(K, a)$. (Infinite $\mathcal{S}$: the same, provided the sum converges.)

Proof. Node balance at state 0: $\lambda_0 p_0 = \mu_1 p_1$. Node balance at state $k \geq 1$: $(\lambda_k + \mu_k) p_k = \lambda_{k-1} p_{k-1} + \mu_{k+1} p_{k+1}$. Assume inductively that $\lambda_{k-1} p_{k-1} = \mu_k p_k$; subtracting it from the node equation leaves $\lambda_k p_k = \mu_{k+1} p_{k+1}$: the induction closes, and each arc balance equation says that across the cut between k and $k+1$ the long-run flow to the right equals the flow to the left. Iterating,

$$p_k = \frac{\lambda_{k-1}}{\mu_k} p_{k-1} = \dots = p_0 \prod_{i=1}^k \frac{\lambda_{i-1}}{\mu_i},$$

and normalisation fixes p_0 . For the switch, $\prod_{i=1}^k \frac{\lambda}{i\mu} = \frac{a^k}{k!}$. □

Remark (time reversibility). Arc balance $p_i q_{ij} = p_j q_{ji}$ for all pairs i, j is stronger than node balance; it holds for birth & death chains and, more generally, for time-reversible chains (Ch. 8, Kolmogorov's criterion). On a graph with a cycle and a chord (CP7 #4) it fails — use node balance there.

(verified in notebook: CP8 #3 (recursion vs solver; Erlang B vs simulated loss), #9)

Checkpoint — answer before moving on.

- **(final-style)** $M/M/2/K$: Poisson arrivals at rate λ , two $\exp(\mu)$ servers, at most K customers in the system. Write λ_k , μ_k , the product form, and the fraction of arrivals lost. Then set $K = \infty$ and state the condition for a limiting distribution to exist.
- For the amoebae (CP7 #5) with λ_0 , $\lambda_k = k\lambda$, $\mu_k = k\mu$ ($k \geq 1$), compute $\prod_{i \leq k} \lambda_{i-1}/\mu_i$ and decide when $\sum_k$ converges. What does divergence mean physically?

- Why is the fraction of *calls* lost equal to the fraction of *time* the switch is full? (Which property of Poisson arrivals is needed? Name it; Ch. 8 proves it.)

AI study card — paste one of these into your AI assistant:

- Make me derive the arc balance equations of a birth-death process from node balance by induction, one state at a time, and then the product-form solution. Then give me an M/M/2/K with numbers and check my limiting distribution.
- Give me a 3-state chain with a cycle 0→1→2→0 at rate 1 and a reverse arc 2→1 at rate 1. Ask me whether arc balance can hold, and make me find p by node balance instead.

Proof 8 — Long-run cost per unit time, and one vs two repair persons

Theorem. Let the CTMC be ergodic and let cost accrue at rate $c(j)$ while in state j . Then $\lim_{T \rightarrow \infty} \frac{1}{T} \int_0^T c(X(t)) dt = \sum_{j \in \mathcal{S}} c(j) p_j$ (almost surely, and in expectation).

Proof. Since $c(X(t)) = \sum_j c(j) \mathbf{1}\{X(t) = j\}$,

$$\frac{1}{T} \int_0^T c(X(t)) dt = \sum_{j \in \mathcal{S}} c(j) \frac{V_j(T)}{T}, \quad V_j(T) = \int_0^T \mathbf{1}\{X(t) = j\} dt,$$

and by Proof 6(c), $V_j(T)/T \rightarrow p_j$ for every j ; a finite sum of convergent terms converges to $\sum_j c(j)p_j$. (For countable $\mathcal{S}$ with $\sum_j |c(j)|p_j < \infty$ the same holds by dominated convergence.) $\square$

Application (CP8 #9). Two identical machines, up time $\exp(\mu)$, repair time $\exp(\lambda)$, revenue r per up machine, c per working repair person. With $X(t)$ = number up:

	one repair person	two repair persons
arc balance	$\lambda p_0 = \mu p_1, \lambda p_1 = 2\mu p_2$	$2\lambda p_0 = \mu p_1, \lambda p_1 = 2\mu p_2$
$\mathbf{p}$	$(2\mu^2, 2\lambda\mu, \lambda^2)/(2\mu^2 + 2\lambda\mu + \lambda^2)$	$(\mu^2, 2\lambda\mu, \lambda^2)/(\mu + \lambda)^2$
reward rates ($j = 0, 1, 2$)	$(-c, r - c, 2r)$	$(-2c, r - c, 2r)$
	N	N
reward per unit time	$\frac{N}{2\mu^2 + 2\lambda\mu + \lambda^2}$	$\frac{N}{(\mu + \lambda)^2}$

with the common numerator $N = -2\mu^2 c + 2\lambda\mu(r - c) + 2r\lambda^2 = 2\lambda\mu(\mu + \lambda)(\frac{r}{\mu} - \frac{c}{\lambda})$. The second denominator is smaller, so two repair persons are better iff $N > 0$, i.e. iff $r/\mu > c/\lambda$: the revenue of one up-period exceeds the cost of one repair. At equality both rewards are 0.

(verified in notebook: CP8 #8 (cost 1,800 in every state), #9 (dashboard: the curves cross at $c = r\lambda/\mu$))

Checkpoint — answer before moving on.

- CP8 #8: the cost is 1,800 won per unit time in *every* state. Why is $\mathbf{p}$ not needed? What if the cost were 1,800 only in state 0?
- **(final-style)** A shop earns r per served customer; use the barbershop chain (CP8 #6) to write the long-run revenue rate as $\sum_j c(j)p_j$ — what is $c(j)$ for a state with busy set j ?
- Why must the cost formula use $\mathbf{p}$ and not the embedded π ? Construct a 2-state example where using π gets the sign of a decision wrong.

AI study card — paste one of these into your AI assistant:

- Quiz me on why the long-run average cost of a CTMC is $\sum_j c(j) p_j$ and not $\sum_j c(j) \pi_j$. Then give me a maintenance problem with numbers and make me decide between two policies.
- Take the two-machine repair problem (rates λ , μ , revenue r , charge c). Ask me to derive the condition under which a second repair person pays off, and check every algebra step of mine.

Where to next. CP8 wrap-up $\rightarrow$ these proofs (gap-fill first) $\rightarrow$ Checkpoint quiz at the start of the next lecture $\rightarrow$ HW3 Part A (closed-book style, no AI on the first attempt) $\rightarrow$ Part C: break the exponential assumption and see which answers survive.